

Pandas is a Python library used to store, organize, clean, and analyze data in table format

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil)
```

```
import pandas as pd
import numpy as np
```

```
print(pd.__version__)
```

```
2.2.2
```

A Pandas Series is a one-dimensional labeled array used to store a single column of data in pandas.

```
#from list
ser=pd.Series([1,2,3,4,5])
print(ser)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

Double-click (or enter) to edit

```
#custom index
ser=pd.Series([1,2,3,4,5], index=['a','b','c','d','e'])
print(ser)
```

```
a    1
b    2
c    3
d    4
e    5
dtype: int64
```

```
student={
    "name":"sireesha",
    "branch":"aiml"
}
se=pd.Series(student)
print(se)
```

```
name      sireesha
branch    aiml
dtype: object
```

```
A=pd.Series([5,6,9,8,7])
```

```
A+2
```

	0
0	7
1	8
2	11
3	10
4	9

```
dtype: int64
```

```
A*3
```

	0
0	15
1	18
2	27
3	24
4	21

```
dtype: int64
```

```
A.mean()
```

```
np.float64(7.0)
```

```
A.sum()
```

```
np.int64(35)
```

```
A.max()
```

```
9
```

```
A.describe()
```

```

      0
count  5.000000
mean   7.000000
std    1.581139
min    5.000000
25%    6.000000
50%    7.000000
75%    8.000000
max    9.000000

```

dtype: float64

```

# Boolean indexing
A[A > 3]

```

```

      0
0  5
1  6
2  9
3  8
4  7

```

dtype: int64

```

# Vectorized string operations
names = pd.Series(['Siree', 'suma', 'renu'])
names.str.lower()
names.str.len()
names.str.startswith('S')

```

```

      0
0  True
1  False
2  False

```

dtype: bool

A DataFrame is a two-dimensional table in pandas used to store data in rows and columns.

```

# From dictionary
student = {
    'Name':  ['siree', 'suma', 'renu', 'iqra'],
    'Age':   [23, 25, 22, 28],
    'Score': [88, 75, 92, 81],
    'Grade': ['A', 'B', 'A', 'A']
}

```

```
a = pd.DataFrame(student)
print(a)
```

	Name	Age	Score	Grade
0	siree	23	88	A
1	suma	25	75	B
2	renu	22	92	A
3	iqra	28	81	A

```
student = [{'name': 'siree', 'age': 23}, {'names': 'devika', 'age': 18}]
b = pd.DataFrame(student)
print(b)
```

	name	age	names
0	siree	23	NaN
1	NaN	18	devika

```
# From NumPy array
arr = np.random.rand(4, 4)
c = pd.DataFrame(arr, columns=['A', 'B', 'C', 'D'], index=['x', 'y', 'z', 'u'])
print(c)
```

	A	B	C	D
x	0.866933	0.716423	0.446543	0.564610
y	0.019147	0.265124	0.648450	0.935749
z	0.870382	0.285225	0.583299	0.385580
u	0.624393	0.133601	0.429722	0.976632

```
# Exporing Data Frame
data = {
    'Name': ['siree', 'renu', 'suma', 'iqra'],
    'Age': [23, 25, 22, 28],
    'Score': [95, 82, 92, 81],
    'Grade': ['A', 'B', 'A', 'A']
}
d = pd.DataFrame(data)
```

```
print(d.head(3))
print(d.tail(3))
print(d.shape)
print(d.info())
print(d.describe())
print(d.columns)
print(d.index)
print(d.dtypes)
print(d.nunique())
print(d.isnull().sum())
```

	Name	Age	Score	Grade
0	siree	23	95	A
1	renu	25	82	B
2	suma	22	92	A

	Name	Age	Score	Grade
1	renu	25	82	B
2	suma	22	92	A
3	iqra	28	81	A

```
(4, 4)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
#   Column  Non-Null Count  Dtype

```

```

---  -----
0   Name    4 non-null    object
1   Age     4 non-null    int64
2   Score   4 non-null    int64
3   Grade   4 non-null    object
dtypes: int64(2), object(2)
memory usage: 260.0+ bytes
None
      Age      Score
count  4.000000  4.000000
mean   24.500000  87.500000
std     2.645751   7.047458
min    22.000000  81.000000
25%    22.750000  81.750000
50%    24.000000  87.000000
75%    25.750000  92.750000
max    28.000000  95.000000
Index(['Name', 'Age', 'Score', 'Grade'], dtype='object')
RangeIndex(start=0, stop=4, step=1)
Name      object
Age       int64
Score     int64
Grade     object
dtype: object
Name      4
Age       4
Score     4
Grade     2
dtype: int64
Name      0
Age       0
Score     0
Grade     0
dtype: int64

```

```
d['Name']
```

	Name
0	siree
1	renu
2	suma
3	iqra

dtype: object

```
d.Name
```

	Name
0	siree
1	renu
2	suma
3	iqra

dtype: object

```
d[['Name', 'Score']]
```

	Name	Score	
0	siree	95	
1	renu	82	
2	suma	92	
3	iqra	81	

```
d.loc[0]
```

0	
Name	siree
Age	23
Score	95
Grade	A

dtype: object

```
d.loc[0:2]
```

	Name	Age	Score	Grade	
0	siree	23	95	A	
1	renu	25	82	B	
2	suma	22	92	A	

```
d.loc[0:2, 'Name':'Score']
```

	Name	Age	Score	
0	siree	23	95	
1	renu	25	82	
2	suma	22	92	

```
print(d.iloc[0])
print(d.iloc[0:2])
print(d.iloc[0:2 , 0:2])
print(d.iloc[-1])
```

```
Name      siree
Age        23
Score      95
Grade      A
Name: 0, dtype: object
   Name  Age  Score  Grade
0  siree  23     95     A
1  renu   25     82     B
   Name  Age
0  siree  23
```

```
1  renu  25
   Name   iqra
   Age    28
   Score   81
   Grade   A
Name: 3, dtype: object
```

```
d.loc[0, 'Name']
```

```
'siree'
```

```
d.iloc[0, 0]
```

```
'siree'
```

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